

In machine learning, feature learning is the process of automatically discovering useful representations from data [6]. It replaces the traditional reliance on features designed by domain experts and is widely believed to underlie the success of modern models such as neural networks [7]. Yet we still lack a mechanistic understanding of when and why these models produce low-dimensional predictive features and what structure those features capture.

Feature learning has two defining aspects: *composition*, through a predictor $g \circ h$, and *joint optimization*, through the simultaneous learning of the feature map h and predictor g . We study this mechanism in a *minimal* form,

$$x \mapsto f(Ux),$$

where $U \in \mathbf{R}^{d \times d}$ is a learned linear feature map and f is a nonlinear function from a reproducing kernel Hilbert space H , e.g., a Sobolev space. Given $X \in \mathbf{R}^d$ and $Y \in \mathbf{R}$, we introduce the following variational problem [1]:

$$\min_U \min_f \mathbf{E}[(Y - f(UX))^2] + \lambda \|f\|_H^2.$$

Fixing $U = I$ recovers classical kernel ridge regression, so the single additional component U *isolates the role of feature learning*. Our central question—understanding what predictive structure in the data distribution U recovers—is the analogue to understanding what the hidden layer weights learn in a 2-layer neural network.

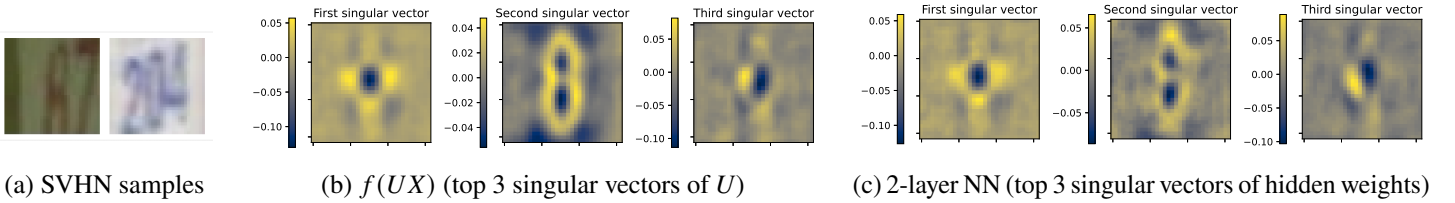


Figure 1: On SVHN, the $f(UX)$ model learns nontrivial representations, as does a 2-layer neural network [3, Section 10].

Although deceptively simple, the $f(UX)$ problem is *genuinely nonlinear and nonconvex* in the learned representation U , as is its 2-layer neural network analogue. At the same time, we find—perhaps surprisingly—that this variational problem *possesses rich mathematical structure*, which allows us to *uncover previously unrecognized feature learning mechanisms*.

We develop our theory in stages, *varying one ingredient at a time*. We begin with a variational foundation of the population problem, characterizing what features of the data distribution are learned by the local minima in U . We discover these minima are connected to variable-selection and scale-detection tasks, and are sensitive to the underlying discrete structure or clustering behavior of the data [1]. We next turn from statics to dynamics, constructing a Riemannian gradient flow on U and showing that in the presence of Gaussian noise variables, the flow possesses infinitely many monotone quantities, so that the noise effect decays automatically [2]. We then pass from the population setting to finite samples and show, perhaps surprisingly, that this denoising persists: empirical minimizers remain low-rank *without explicit rank or norm regularization on U* [3]. Finally, restricting U to be diagonal reveals how the choice of RKHS governs feature recovery: local minima associated with the Laplace RKHS H recover all relevant variables, whereas those with the Gaussian RKHS H need not [4]. These results show how noise suppression, scale detection and low-rank structure emerge without traditional explicit structural regularization.

These results are significant in two respects. First, to our knowledge, analogous phenomena remain challenging to establish for neural networks at comparable generality [2, Postscript]. Second, in the theory of neural nets, architecture, optimization dynamics, finite-sample effects, and regularization are often entangled, making individual roles difficult to identify. Our framework, by contrast, *isolates these mechanisms one at a time*, thus revealing how each contributes to feature learning.

These results point to two complementary directions for future research. First, the emerging theory raises a broader question: which mechanisms of feature learning are *universal* consequences of *composition* and *joint optimization*, and which are induced by specific architectural choices? Viewing $f(Ux)$ as a basic model of a neural-network layer, a natural next step is to determine whether noise suppression, scale detection, and feature recovery persist in neural networks [5] or rely on kernel-specific structure. Second, motivated by the state-of-the-art empirical success of feature-learning kernel models on tabular datasets [8, 9], a complementary direction is to develop computationally scalable methods for jointly optimizing $f(UX)$ by combining random-feature approximations of f with neural-network optimization techniques. Together, these directions form a two-way bridge: kernel learning $f(UX)$ provides a tractable theory for understanding feature learning, with potential implications for neural nets, while neural-net methods make kernel feature learning $f(UX)$ practical at scale.

References

Selected work by the group

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Selected related work

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